Learner Full Name: _____

School: _____ **Class:** _____

Workstation Device: _____ **Station #:** _____

Session 1: Meet Your Computer

Welcome! Whether you are sitting in front of a **Desktop computer** or a **Laptop**, this workbook will guide you step-by-step. You do not need any prior experience.

1. What is a Computer? (The Corn Mill Analogy)

Think about a local neighborhood corn mill:

- **1. INPUT:** You pour raw, dry corn into the funnel hopper.
- **2. PROCESS:** The motor grinds and crushes the grains inside the machine housing.
- **3. OUTPUT:** Clean, fine corn meal pours into your waiting basin.

Every computer—desktop or laptop—works the exact same way:

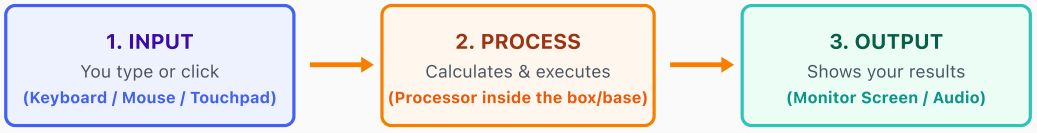


Figure 1.0: The flow of computing. Input becomes Output through internal processing.

2. Which Machine Are You Using Today?

🖥️ OPTION A: Desktop Computer

A stationary machine designed to stay on a table. It is made of **separate pieces** connected with cables: a monitor screen, a system unit tower, an external keyboard, and a mouse.

💻 OPTION B: Laptop Computer

A portable all-in-one computer. The screen folds down over the keyboard. It has an internal rechargeable battery, a built-in keyboard, and a flat **touchpad** that acts as the mouse.

3. Hardware vs. Software in 30 Seconds

Hardware (The Physical Body)	Software (The Brain's Programs)
Anything you can physically **touch with your hand**.	The invisible instructions that tell the machine **what to do**.
Examples: Screen glass, keys, mouse, touchpad, battery.	*Examples:* Word processor, Scratch coding, games, web browser.

HARDWARE ANATOMY: DESKTOP VS. LAPTOP

1. Comparing the Physical Machines

Study the two setups below to understand where each part lives on your machine:

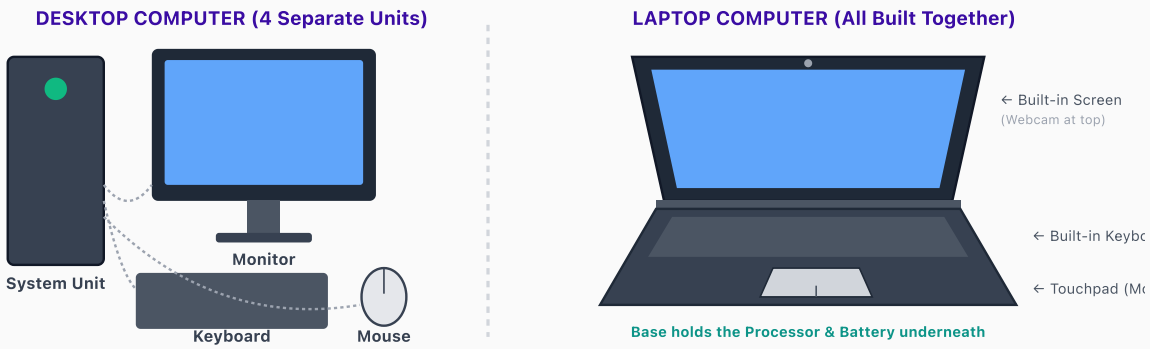


Figure 1.1: Visual comparison of Desktop vs. Laptop. Both contain the exact same core computing parts!

2. Hardware Component Location Table

Component	On a Desktop	On a Laptop
The "Brain" (CPU)	Inside the heavy metal System Unit tower.	Packed flat underneath the built-in keyboard.
The Display Screen	A standalone glass monitor on a heavy stand.	Hinged top lid that folds open and closed.
The Pointer Tool	An external handheld plastic **Mouse**.	A built-in touch-sensitive **Touchpad**.
Power Source	Constantly plugged into wall / UPS battery.	Rechargeable battery (charges with an adapter).

3. Critical Care Rules for Laptops

⚠ **THREE GOLDEN LAPTOP CARE RULES**

1. **Open from the center:** Always open the laptop lid by lifting gently from the center of the rim. Never yank it open from just one corner, or the hinge may crack!
2. **Two hands at all times:** When carrying a laptop, close the screen completely and carry it with **two hands** close to your chest like a valuable textbook. Never hold it by its screen!
3. **Keep vents clear:** Never place a running laptop on a soft mattress, pillow, or cloth. It breathes air through vents on the bottom. Always use a flat, hard desk so it does not overheat.

SAFE STARTUP: DESKTOP VS. LAPTOP

1. Choose Your Machine's Startup Pathway

Follow the column that matches the machine you are sitting in front of today:

🖥️ If You Have a Desktop:

1. **Wall Switch & UPS:** Check that the wall switch and the black **UPS** box are ON (green light). Your mentor switches these on.
2. **System Unit Power:** Press the large circular power button **[(I)]** on the front of the System Unit once. Listen for the fan breeze.
3. **Monitor Power:** Look at the bottom-right rim of the monitor. If the small power light is off, press the monitor power button once.
4. **Wait to Boot:** Keep your hands off. Wait 30–60 seconds for the Desktop screen to load.

💻 If You Have a Laptop:

1. **Open the Lid Gently:** Hold the base down with one hand. Use your other hand to lift the lid from the center until it forms an open angle.
2. **Check Power Cord:** If the battery is low, ensure the charging adapter cable is plugged into the small side charging port and wall socket.
3. **Find Laptop Power Key:** Look at the top-right corner of the built-in keyboard (or the side edge). Press the small power button **once firmly**.
4. **Wait to Boot:** Do not touch any keys. Wait for the screen to light up and show the Welcome screen.

2. Workstation Ergonomics (Both Setups)



Figure 1.2: Ergonomic posture tips for both machines. Keep your back straight!

3. Startup Troubleshooting ("What Went Wrong?")

🚨 MY MACHINE IS NOT TURNING ON!

- **If using a Desktop:** Check the monitor power light first. If the System Unit fan is dead silent, check whether the UPS or wall socket is switched ON.
- **If using a Laptop:** The battery might be completely flat ("dead"). Plug in the charging cord and look for a tiny amber/white charging LED on the side edge. Wait 2 minutes, then press the power button again.

POINTER CONTROL: MOUSE VS. TOUCHPAD

1. Controlling the Pointer (Two Ways)

To control the arrow on screen, Desktop users use a **Mouse**, while Laptop users use a built-in **Touchpad** (also called a Trackpad). Both achieve the exact same results!

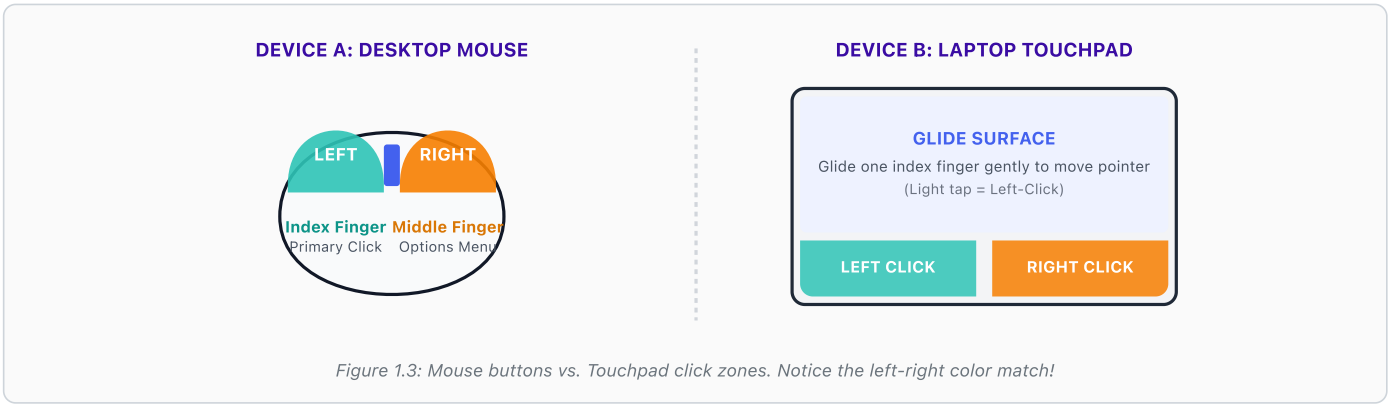


Figure 1.3: Mouse buttons vs. Touchpad click zones. Notice the left-right color match!

2. Master Action Reference Guide

Action	🖥️ How to do it with a MOUSE	💻 How to do it on a TOUCHPAD
Move Pointer	Slide the mouse across the pad.	Glide **one finger** lightly across the touchpad.
Left-Click (Select)	Tap the **Left Button** once with your index finger.	Click the **bottom-left corner** firmly, OR gently tap the pad surface once.
Double-Click (Open)	Tap the Left Button twice fast: *Click-Click!*	Tap your finger twice quickly on the pad surface.
Right-Click (Menu)	Tap the **Right Button** with your middle finger.	Click the **bottom-right corner** firmly with one finger.
Scroll Page	Roll the center plastic wheel forward/back.	Place **two fingers together** and slide them up or down simultaneously.

3. Screen Cursor Shapes

Look closely at your screen: The pointer changes shape to tell you what it is doing!

- **White Arrow:** Normal state, ready to point and select.
- **Hand with Pointing Finger:** You are hovering over a website link or clickable button.
- **"I" Bar (I-Beam):** You are over a text area; click here to start typing words.
- **Spinning Blue Circle:** Hold on! The computer is busy thinking. Do not click.

1. Guided Practical Lab: Your First Screen Actions

Complete these 3 foundational challenges at your workstation:

A **Challenge 1: The Screen Orbit**

Mouse Users: Slide the mouse until the arrow reaches all 4 corners of your screen. If you run out of desk space, lift the mouse in the air, put it back in the middle of the pad, and glide again.

Touchpad Users: Practice gliding one index finger to each corner. If you reach the edge of the touchpad, lift your finger, place it in the center, and glide again!

B **Challenge 2: Icon Selection Practice**

Move your arrow over the **Recycle Bin** icon on the desktop.

Perform a **single Left-Click** (or tap once on your touchpad).

Observation: Notice how the icon is highlighted in blue. Now click any empty space on the wallpaper to deselect it.

C **Challenge 3: Reveal the Secret Options Menu**

Move the pointer to an empty space on the desktop. Perform a **Right-Click** (Right mouse button OR bottom-right touchpad corner).

Observation: A menu pops up! Tap the [Esc] key on your keyboard to close it safely.

2. Safe Software Shutdown (The Official Pathway)

⚠ **UNIVERSAL RULE FOR BOTH DESKTOPS & LAPTOPS**

Never turn off a desktop at the wall, and **never** hold down the laptop power button to force it off! Doing so cuts power while the computer is storing files. Always follow the 4-step software shutdown:

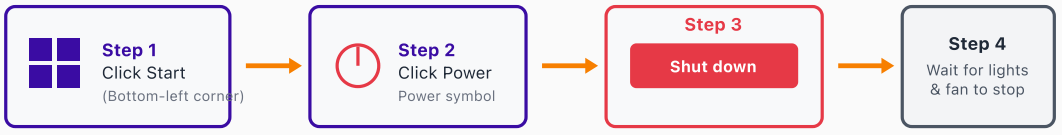


Figure 1.4: Universal 4-Step Software Shutdown Process.

3. Departure Routine Checklist

🖥 If You Used a Desktop:

- [ ] The screen is black and the fan has stopped.
- [ ] I turned off the monitor's power button.
- [ ] Keyboard and mouse parked neatly on the desk.

💻 If You Used a Laptop:

- [ ] The screen is black and all side lights are off.
- [ ] I gently closed the lid using two hands from center.
- [ ] Charger cord neatly coiled (if unplugged).

SESSION 1 REVIEW & VERIFICATION

1. Vocabulary Match-Up (Draw a Line)

Draw a line connecting each computer term on the left to its correct job on the right:

[System Unit / Base]	• Touch-sensitive surface on a laptop that works like a mouse.
[Touchpad]	• Battery backup that prevents computers from cutting out during power outages.
[Monitor]	• The main processing unit containing the computer's electronic "brain".
[UPS]	• Displays documents, pictures, and the desktop on its screen.
[Right-Click]	• Action that opens a secret options menu for the selected item.

2. Written Review Questions

Question 1: Where is the "brain" (CPU) located on a laptop computer?

Question 2: Why must you open a laptop lid gently from the center rather than pulling one corner?

Question 3: You slide your mouse (or finger on the touchpad) and the cursor arrow reaches the far edge of the pad, but you still need it to travel further on screen. What physical movement do you do?

3. Self-Assessment & Mentor Sign-Off

Tick the box that honestly reflects your skill level today:

Competency Demonstrated	With Help	Almost Alone	All By Myself!
1. I can identify whether my station is a Desktop or Laptop and name its parts.	○	○	○
2. I can start up my machine safely following the correct sequence.	○	○	○
3. I can accurately Left-Click and Right-Click using the Mouse or Touchpad.	○	○	○
4. I can perform a safe software shutdown via the Start Menu.	○	○	○

Mentor Signature: _____ **Date Completed:** _____